KOM4521 Multivariable Control Theory

2024-2025 Fall

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Week 4-5-6

- Diagonalization, Solution of LTI Systems
- Stability of LTI Systems, Lyapunov Stability Theorem
- Controllability and Observability

Transformations:

- State-space representations are not unique. Selection of vector components in x is quite arbitrary.

Analyze the tranformation

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t).$$

Let $x(t) = Tz(t)$,where T is an invertible (similarity) transformation matrix.

Transformations:

$$\begin{aligned}
\dot{z}(t) &= T^{-1}\dot{x}(t) \\
&= T^{-1}[Ax(t) + Bu(t)] \\
&= T^{-1}[ATz(t) + Bu(t)] \\
&= \underbrace{T^{-1}AT}_{\bar{A}} z(t) + \underbrace{T^{-1}B}_{\bar{B}} u(t) \\
y(t) &= \underbrace{CT}_{\bar{C}} z(t) + \underbrace{D}_{\bar{D}} u(t)
\end{aligned}$$

so
$$\begin{aligned}
\dot{z}(t) &= \bar{A}z(t) + \bar{B}u(t) \\
y(t) &= \bar{C}z(t) + \bar{D}u(t).
\end{aligned}$$

Transformations:

$$H_1(s) = C(sI - A)^{-1}B + D$$

$$H_2(s) = \bar{C}(sI - \bar{A})^{-1}\bar{B} + \bar{D}.$$

- Need $H_1(s) = H_2(s)$

$$\begin{aligned} H_1(s) &= C(sI - A)^{-1}B + D \\ &= CTT^{-1}(sI - A)^{-1}TT^{-1}B + D \\ &= (CT)[T^{-1}(sI - A)T]^{-1}(T^{-1}B) + D \\ &= \bar{C}(sI - \bar{A})^{-1}\bar{B} + \bar{D} = H_2(s). \end{aligned}$$

Transfer function is not changed by similarity transform.

Transformations:

Example: Controller canonical form for

$$\frac{2s+3}{(s+1)(s+2)} = \frac{2s+3}{s^2+3s+2}$$

$$\dot{x}_c(t) = \begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix} x_c(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 2 & 3 \end{bmatrix} x_c(t).$$

Transformations:

Suppose we have the transformation matrix $\det(T) = 1$ (invertible).

$$T = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}.$$

- Let $x_c = T\bar{x}$, where $\bar{x}$ is a new state.

$$\dot{\bar{x}}(t) = (T^{-1}AT)\bar{x}(t) + (T^{-1}B)u(t)$$

$$y(t) = (CT)\bar{x}(t).$$

Transformations:

Plugging in A, B, C, T

$$\dot{\bar{x}}(t) = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix} \bar{x}(t) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 1 \end{bmatrix} \bar{x}(t),$$

which gives the diagonal realization of the transfer function!

■ We'll often change coordinates in a system, for example to solve a particular problem more easily.

Time Response:

Homogenous part(scalar):

- $\dot{x}(t) = a\,x(t), \qquad\qquad x(0)$
- Take Laplace

  $\mathrm{X}(s) = (s-a)^{-1}x(0).$
- Inverse Laplace

  $x(t) = e^{at}x(0).$

Time Response:

Homogenous part(full):

- $\dot{x}(t) = A\,x(t), \qquad x(0)$
- Take Laplace

$$X(s) = (sI - A)^{-1}x(0).$$

- Inverse Laplace

$$(sI - A)^{-1} = \frac{I}{s} + \frac{A}{s^2} + \frac{A^2}{s^3} + \cdots$$

$$\mathcal{L}^{-1}[(sI - A)^{-1}] = I + At + \frac{A^2t^2}{2!} + \frac{A^3t^3}{3!} + \cdots$$

$$\triangleq e^{At} \qquad \text{matrix exponential}$$

$$x(t) = e^{At}x(0).$$

Time Response:

- e^{At} is called the transition matrix or **state-transition** matrix.

- $e^{(A+B)t} = e^{At}\ e^{Bt}$ iff $AB = BA$

Time Response:

Example:

$$\dot{x} = Ax, \qquad A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

$$(sI - A)^{-1} = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}^{-1} = \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \frac{1}{(s+2)(s+1)}$$

$$= \begin{bmatrix} \dfrac{2}{s+1} - \dfrac{1}{s+2} & \dfrac{1}{s+1} - \dfrac{1}{s+2} \\ \dfrac{-2}{s+1} + \dfrac{2}{s+2} & \dfrac{-1}{s+1} + \dfrac{2}{s+2} \end{bmatrix}$$

$$e^{At} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} 1(t)$$

Time Response:

Forced solution(scalar):

$$\dot{x}(t) = ax(t) + bu(t), \quad x(0)$$

$$x(t) = e^{at}x(0) + \underbrace{\int_0^t e^{a(t-\tau)}bu(\tau)\,\mathrm{d}\tau}_{\text{convolution}}.$$

1. $\dot{x}(t) - ax(t) = bu(t)$
2. $e^{-at}[\dot{x}(t) - ax(t)] = \dfrac{\mathrm{d}}{\mathrm{d}t}[e^{-at}x(t)] = e^{-at}bu(t).$
3. $\int_0^t \dfrac{\mathrm{d}}{\mathrm{d}\tau}[e^{-a\tau}x(\tau)]\,\mathrm{d}\tau = e^{-at}x(t) - x(0) = \int_0^t e^{-a\tau}bu(\tau)\,\mathrm{d}\tau.$

Time Response:

Forced solution(full):

$$\boxed{x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)\,\mathrm{d}\tau}$$

$$y(t) = Cx(t) + Du(t),$$

$$y(t) = \underbrace{Ce^{At}x(0)}_{\text{initial resp.}} + \underbrace{\int_0^t Ce^{A(t-\tau)}Bu(\tau)\,\mathrm{d}\tau}_{\text{convolution}} + \underbrace{Du(t)}_{\text{feedthrough}}.$$

Diagonalization:

If the matrix A is diagonalizable, that is, if there exists a matrix T such that $T^{-1}AT = \Lambda$ diagonal.

$e^{At} = Te^{\Lambda t}T^{-1}$, and

$$e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & & & 0 \\ & e^{\lambda_2 t} & & \\ & & \ddots & \\ 0 & & & e^{\lambda_n t} \end{bmatrix}.$$

Diagonalization:

Assume that eigenvectors $v_1, v_2, .., v_n$ are linearly independent.

Stack together in a matrix equation:

$$Av_i = \lambda_i v_i \qquad i = 1, 2, \ldots, n$$

$$A \underbrace{\begin{bmatrix} v_1 & v_2 & \ldots & v_n \end{bmatrix}}_{T} = \begin{bmatrix} v_1 & v_2 & \ldots & v_n \end{bmatrix} \underbrace{\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}}_{\Lambda}$$

$$AT = T\Lambda \quad \Rightarrow \quad T^{-1}AT = V^{-1}AV = \Lambda.$$

Diagonalization:

- We have found the transformation matrix T that diagonalizes A!
- However, not all matrices are diagonalizable. For example,

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \qquad \det(\lambda I - A) = \lambda^2.$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_a \\ v_b \end{bmatrix} = 0 \quad \Rightarrow \quad \text{all vectors of the form } \begin{bmatrix} v_a \\ 0 \end{bmatrix} \neq 0.$$

Diagonalization:

- Diagonal form makes it easier to understand what is happening in a state-space dynamic system.
  - Assume A is diagonalizable by $T = V$.
  - Define new coordinates by $x(t) = T\tilde{x}(t)$ so

$$\dot{\tilde{x}}(t) = T^{-1}Ax(t) = T^{-1}AT\tilde{x}(t) = \Lambda\tilde{x}(t).$$

  - In new coordinate system, system is diagonal (decoupled).

Diagonalization:

- Trajectories consist of n independent modes; that is,

$$\tilde{x}_i(t) = e^{\lambda_i t}\tilde{x}_i(0)$$

hence the name modal form.

- To understand in original coordinate system, write $T^{-1}AT = \Lambda$ as $T^{-1}A = \Lambda T^{-1}$ with

$$T^{-1} = \begin{bmatrix} w_1^T \\ w_2^T \\ \vdots \\ w_n^T \end{bmatrix},$$

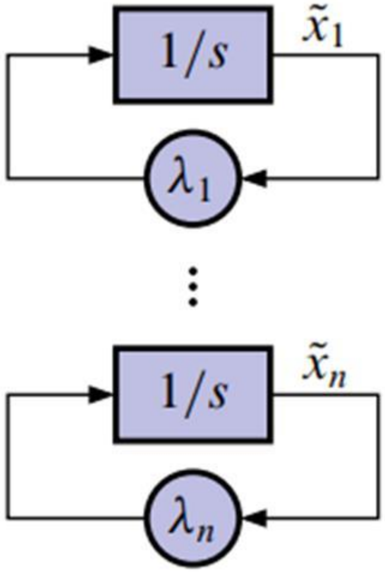
i.e., rows of T^{-1}.

ω_i is a left eigenvector of A.

Diagonalization:

$$e^{At} = Te^{\Lambda t}T^{-1}$$

$$= \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix} \begin{bmatrix} e^{\lambda_1 t} & & & 0 \\ & e^{\lambda_2 t} & & \\ & & \ddots & \\ 0 & & & e^{\lambda_n t} \end{bmatrix} \begin{bmatrix} w_1^T \\ w_2^T \\ \vdots \\ w_{nT} \end{bmatrix} = \sum_{i=1}^{n} e^{\lambda_i t} v_i w_i^T .$$



$$\begin{aligned} x(t) &= e^{At}x(0) \\ &= Te^{\Lambda t}T^{-1}x(0) \\ &= \sum_{i=1}^{n} e^{\lambda_i t} v_i (w_i^T x(0)). \end{aligned}$$

Jordan Canonical Form:

- Any matrix $A \in \mathbb{R}^{nxn}$ can be put in Jordan canonical form by a similarity transformation.
- That is, we can find a transformation matrix T such that $T^{-1}AT = J$,where J is a matrix in Jordan form.

Jordan Canonical Form:

- Jordan canonical form of a matrix is block-diagonal and looks like

$$J = \begin{bmatrix} J_1 & & 0 \\ & \ddots & \\ 0 & & J_q \end{bmatrix}$$

where

$$J_i = \begin{bmatrix} \lambda_i & 1 & & 0 \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_i \end{bmatrix} \in \mathbb{R}^{n_i \times n_i}$$

Jordan Canonical Form:

- J is block-diagonal and upper bidiagonal.
- J is diagonal is the special case of n Jordan blocks of size $n_i = 1$.
- Jordan form is unique (up to permutations of the blocks).

Jordan Canonical Form:

- $\chi(\lambda) = \det(\lambda I - A) = \det(\lambda I - J) = (\lambda - \lambda_1)^{n_1} \cdots (\lambda - \lambda_q)^{n_q} = 0$ hence distinct eigenvalues ➠ $n_i = 1$ ➠ A diagonalizable.
- $\dim \mathcal{N}(\lambda_i I - A) = \dim \mathcal{N}(\lambda_i I - J)$ is the number of Jordan blocks with eigenvalue λ_i.
- The sizes of each Jordan block may also be computed, but this is complicated. *i.e.*, leave it to MATLAB! ➠ `jordan(A)`

Jordan Canonical Form:

Example: Consider

$$A = \begin{bmatrix} 3 & -1 & 1 & 1 & 0 & 0 \\ 1 & 1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}.$$

$$\det(\lambda I - A) = (\lambda - 2)^5 \lambda.$$

- $\operatorname{rank}(2I - A) = 4$, so $\dim(\mathcal{N}(2I - A)) = 6 - 4 = 2$ so there are two Jordan blocks with eigenvalue 2.

Jordan Canonical Form:

- We can check this in MATLAB: `jordan(A)`

$$J = \left[\begin{array}{c|ccc|cc} 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 \end{array}\right]$$

Jordan Canonical Form:

If A is not diagonalizable, the Cayley-Hamilton Theoram is evaluated using the Jordan form and Jordan blocks:

$$A = T^{-1} J T.$$

Consider a sketch of the proof for a Jordan block of size 2 and

$$\chi(\lambda_i) = \lambda_i^3 + a_2 \lambda_i^2 + a_1 \lambda_i + a_0 = 0.$$

Jordan Canonical Form:

Then,

$$J_i = \begin{bmatrix} \lambda_i & 1 \\ 0 & \lambda_i \end{bmatrix}$$

$$\chi(J_i) = \begin{bmatrix} \lambda_i^3 & 3\lambda_i^2 \\ 0 & \lambda_i^3 \end{bmatrix} + a_2 \begin{bmatrix} \lambda_i^2 & 2\lambda_i \\ 0 & \lambda_i^2 \end{bmatrix} + a_1 \begin{bmatrix} \lambda_i & 1 \\ 0 & \lambda_i \end{bmatrix} + a_0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

We can easily see that the diagonal and lower-diagonal components are zero so

$\chi(J_i) = \begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix}$ where $\alpha = 3\lambda_i^2 + 2a_2\lambda_i + a_1$

Jordan Canonical Form:

But, $\alpha = \frac{d}{d\lambda}\chi(\lambda)=0$ which completes the sketch.

The Cayley–Hamilton theorem shows us that A^n is a function of matrix powers A^{n-1} down to A^0. Therefore, to compute any polynomial of A it suffices to compute only powers of Aup to $A^{n-1}$and appropriately weight their sum. A lot of proofs use the Cayley–Hamilton theorem.

Lyapunov Stability:

Consider a continuous-time LTV system

$$\dot{x}(t) = A(t)x(t) + B(t)u(t)$$

$$y(t) = C(t)x(t) + D(t)u(t).$$

- ✓ The system is **(marginally)** stable in the sense of Lyapunov or internally stable if, for every $x(t_0) = x_0$, the homogeneous

$$x(t) = \Phi(t, t_0)x_0, \qquad t \geq 0$$

is <u>uniformly bounded</u> (all trajectories can be bounded by a single constant vector).

Lyapunov Stability:

- ✓ It is additionally **asymptotically** stable if, for every $x(t_0) = x_0, x(t) = 0$ as $t \to \infty$.

- ✓ It is additionally **exponentially** stable if there exist constants c, λ such that for every initial condition $x(t_0) = x_0$

$$\|x(t)\| \leq ce^{\lambda(t-t_0)}\|x_0\|, \qquad t \geq 0.$$

Lyapunov Stability:

- ✓ It is additionally **asymptotically** stable if, for every $x(t_0) = x_0$, $x(t) = 0$ as $t \to \infty$.

- ✓ Finally, it is **unstable** if it is not marginally stable. The state may diverge over time, depending on specific x_0.

Lyapunov Stability:

■ Therefore, a continuous-time LTI system is:

- (Marginally) stable iff all eigenvalues of A have negative or zero real parts and all Jordan blocks for eigenvalues having zero real parts are $1 \ x \ 1$ in dimension.
- Asymptotically stable iff all eigenvalues have negative real parts.

Lyapunov Stability:

- Exponentially stable iff all eigenvalues have negative real parts.
- Unstable iff at least one eigenvalue has pos. real part, or a Jordan block for an eigenvalue having zero real part is bigger than $1\ x\ 1$.

Lyapunov Stability:

When all eigenvalues have negative real part, we can find c, λ so that

$$\|x(t)\| \leq ce^{\lambda(t-t_0)}\|x_0\|$$

Therefore,

$$\|x(t)\| = \|e^{A(t-t_0)}x_0\| \leq \|e^{A(t-t_0)}\|\|x_0\| \leq ce^{-\lambda(t-t_0)}\|x_0\|.$$

Asymptotic and exponential stability are equivalent for LTI systems.

Lyapunov Stability Theorem:

- The eigenvalue test will often be sufficient for evaluating Lyapunov stability, but some extra properties can be helpful.
- The Lyapunov stability theorem provides an alternative condition to check whether a continuous-time homogeneous LTI system is asymptotically stable.

Lyapunov Stability Theorem:

The theorem states that the following five conditions are equivalent:

- The system is asymptotically stable.
- The system is exponentially stable.
- All the eigenvalues of A have strictly negative real parts.
- For every symmetric positive-definite matrix Q, there exists a unique solution P to the following Lyapunov equation

Lyapunov Stability Theorem:

$$A^T P + PA = -Q.$$

Further, P is symmetric and positive-definite.

- There exists a symmetric positive-definite matrix P for which the following Lyapunov matrix inequality holds

$$A^T P + PA < 0.$$

Proof of the Lyapunov Stability Theorem

- We start by showing that the unique solution to the Lyapunov equation $A^T P + PA = -Q$ is

$$P = \int_0^\infty e^{A^T t} Q e^{At} \, \mathrm{d}t .$$

- We verify this in four steps. The first step is to show that the integral is well defined (finite).
  - This is true since the system is exponentially stable, so e^{At} and hence $\|e^{A^T t} Q e^{At}\|$ converges to zero exponentially quickly as $t \to \infty$. Therefore, the integral is absolutely convergent.

Proof of the Lyapunov Stability Theorem

- Second, we show that this integral solves the Lyapunov equation via direct substitution:

$$A^T P + PA = \int_0^\infty A^T e^{A^T t} Q e^{At} + e^{A^T t} Q e^{At} A \, \mathrm{d}t.$$

- But (noting that $e^{At} A = A e^{At}$ via infinite-series expansion of e^{At}),

$$\frac{\mathrm{d}}{\mathrm{d}t}(e^{A^T t} Q e^{At}) = A^T e^{A^T t} Q e^{At} + e^{A^T t} Q e^{At} A.$$

- So,

$$A^T P + PA = \int_0^\infty \frac{\mathrm{d}}{\mathrm{d}t}(e^{A^T t} Q e^{At}) \, \mathrm{d}t = \left[e^{A^T t} Q e^{At} \right|_0^\infty$$

$$= \left(\lim_{t \to \infty} e^{A^T t} Q e^{At} \right) - e^{A^T 0} Q e^{A0}.$$

Proof of the Lyapunov Stability Theorem

- Because of asymptotic stability, the first term vanishes.
- Because $e^{A^T 0} = I$, the second term becomes $-Q$.
- So, the proposed solution solves the equation, but we don't yet know whether it is symmetric, positive-definite, and/or unique.

- Third, to show that it is symmetric we compute

$$\begin{aligned} P^T &= \int_0^\infty \left(e^{A^T t} Q e^{At}\right)^T \mathrm{d}t \\ &= \int_0^\infty \left(e^{At}\right)^T Q^T \left(e^{A^T t}\right)^T \mathrm{d}t \\ &= \int_0^\infty e^{A^T t} Q e^{At} \,\mathrm{d}t = P. \end{aligned}$$

Proof of the Lyapunov Stability Theorem

- To show that it is positive-definite, we pick an arbitrary constant vector $z \neq 0$ and compute

$$\begin{aligned} z^T P z &= \int_0^\infty z^T \left(e^{A^T t} Q e^{At} \right) z \, \mathrm{d}t \\ &= \int_0^\infty w(t)^T Q w(t) \, \mathrm{d}t, \end{aligned}$$

  which has positive-definite form since Q is positive definite.
- Further, we see that $z^T P z = 0$ only when $w(t) = e^{At} z = 0$, which can only happen when $z = 0$ for finite time.
- Therefore P is positive-definite.

- Fourth, to see that no other matrix solves this equation, we proceed using a proof by contradiction.

Proof of the Lyapunov Stability Theorem

- That is, assume that there exists another solution $\bar{P}$ to the Lyapunov equation. If this were so, we would have both

$$A^T P + PA = -Q$$

$$A^T \bar{P} + \bar{P} A = -Q.$$

- Then,

$$A^T(P - \bar{P}) + (P - \bar{P})A = 0.$$

- Multiplying on the left by $e^{A^T t}$ and on the right by e^{At}

$$e^{A^T t} A^T (P - \bar{P}) e^{At} + e^{A^T t}(P - \bar{P}) A e^{At} = 0.$$

Proof of the Lyapunov Stability Theorem

- Multiplying on the left by $e^{A^T t}$ and on the right by e^{At}

$$e^{A^T t} A^T (P - \bar{P}) e^{At} + e^{A^T t} (P - \bar{P}) A e^{At} = 0.$$

- On the other hand,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{A^T t}(P - \bar{P})e^{At}\right) = e^{A^T t} A^T (P - \bar{P}) e^{At} + e^{A^T t}(P - \bar{P}) e^{At} A = 0.$$

- Therefore, $e^{A^T t}(P - \bar{P})e^{At}$ must be constant for all time, which requires that $P = \bar{P}$ since e^{At} is nonsingular.

- We have now shown that condition 2 implies 4.
- It immediately follows that 4 implies 5 if we select $Q = I$.
- To show that condition 5 implies condition 2, let P be a symmetric positive-definite matrix that satisfies

$$A^T P + PA < 0$$

and let $Q = -(A^T P + PA) > 0$.

Wind

Proof of the Lyapunov Stability Theorem

- Consider an arbitrary solution $x(t)$ to the homogeneous system dynamics and define the scalar signal

$$v(t) = x^T(t)Px(t) \geq 0.$$

- This signal is a kind of weighted "energy of the state" for the homogeneous system.
- For a stable homogeneous system, we would expect this energy to dissipate over time, which is exactly what we will show.
  - This idea returns when talking about nonlinear stability, later.
- Taking derivatives of $v(t)$,

$$\dot{v}(t) = \dot{x}^T(t)Px(t) + x^T(t)P\dot{x}(t).$$

Proof of the Lyapunov Stability Theorem

- But, since $\dot{x}(t) = Ax(t)$,

$$\dot{v}(t) = x^T(t)\left(A^T P + PA\right)x(t) = -x^T(t)Qx(t) \leq 0.$$

- Therefore $v(t)$ is a nonincreasing signal and we conclude that

$$v(t) = x^T(t)Px(t) \leq v(0) = x^T(0)Px(0), \qquad t \geq 0.$$

- But since $v(t) = x^T(t)Px(t) \geq \lambda_{\min}[P]\|x(t)\|^2$, we conclude that

$$\|x(t)\|_2^2 \leq \frac{x^T(t)Px(t)}{\lambda_{\min}[P]} = \frac{v(t)}{\lambda_{\min}[P]} \leq \frac{v(0)}{\lambda_{\min}[P]}, \quad t \geq 0.$$

- This means that the system is stable. To show that it is exponentially stable, we combine our known results that

$$x^T(t)Qx(t) \geq \lambda_{\min}[Q]\|x(t)\|_2^2$$

and that $v(t) = x^T(t)Px(t) \leq \lambda_{\max}[P]\|x(t)\|_2^2$ and conclude

$$\dot{v}(t) = -x^T(t)Qx(t) \leq -\lambda_{\min}[Q]\|x(t)\|_2^2 \leq -\frac{\lambda_{\min}[Q]}{\lambda_{\max}[P]}v(t),$$

Proof of the Lyapunov Stability Theorem

or that

$$\dot{v}(t) \le \mu v(t).$$

- It can be shown (see text) that this converges more quickly than the exponential

$$v(t) \le e^{\mu(t-t_0)} v(t_0).$$

- Since $v(t)$ converges exponentially quickly, so too must $\|x(t)\|_2$ since we have already shown that

$$\|x(t)\|_2^2 \le \frac{v(t)}{\lambda_{\min}[P]}.$$

- This finishes our proof of the Lyapunov stability theorem for LTI continuous-time systems.

Lyapunov Stability Theorem:

Example: A homogenous system is given below $\dot{x}(t) = Ax(t)$ where $A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}$ find out if the Lyapunov stable.

$$A^T P + P^T A = -Q$$

$Q = I$ symmetric matrix

$P^T = P$ positive definite

Lyapunov Stability Theorem:

$$A^T P + P^T A = -I$$

$$\begin{bmatrix} 0 & -1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} -2p_{12} & p_{11} - 2p_{12} - p_{22} \\ p_{11} - 2p_{12} - p_{22} & 2p_{12} - 4p_{22} \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$p_{12} = 0.5$$
$$p_{22} = 0.5$$
$$p_{11} = 1.5$$

Lyapunov Stability Theorem:

$$A^T P + PA = -Q.$$

Further, P is symmetric and positive-definite.

- There exists a symmetric positive-definite matrix P for which the following Lyapunov matrix inequality holds

$$A^T P + PA < 0.$$

Input–output stability: LTV case

Consider the continuous-time LTV system

$$\dot{x}(t) = A(t)x(t) + B(t)u(t)$$

$$y(t) = C(t)x(t) + D(t)u(t).$$

The forced response of this system in the absence of initial conditions is

$$y(t) = \int_0^t C(t)\Phi(t,\tau)B(\tau)u(\tau)\,\mathrm{d}\tau + D(t)u(t).$$

Input–output stability: LTV case

This system is said to be (uniformly) bounded-input bounded-output (BIBO) stable if there exists a finite constant γ such that, for every input u(t), its forced response satisfies

$$\sup_{t\in[0,\infty)} \|y(t)\| \le \gamma \sup_{t\in[0,\infty)} \|u(t)\|.$$

Time-domain conditions for BIBO stability

The following two statements are equivalent:

1. A LTV system is uniformly BIBO stable.
2. Every entry of the system's $D(t)$ is uniformly bounded and

$$\sup_{t \geq 0} \int_0^t |g_{ij}(t, \tau)| \, d\tau < \infty$$

for every entry $g_{ij}(t, \tau)$ of $C(t)\phi(t,\tau)B(\tau)$.

Input–output stability: LTI case

Consider the continuous-time LTI system

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t),$$

We have that

$$C\Phi(t, \tau)B = Ce^{A(t-\tau)}B.$$

Input–output stability: LTI case

We can therefore rewrite our prior stability criterion as finite D plus

$$\sup_{t \geq 0} \int_0^t |\bar{g}_{ij}(t-\tau)| \, \mathrm{d}\tau < \infty$$

understanding that $g_{ij}(t-\tau)$ denotes the ij th entry of $Ce^{A(t-\tau)}B$. $\rho = t - \tau$

$$\sup_{t \geq 0} \int_0^t |\bar{g}_{ij}(t-\tau)| \, \mathrm{d}\tau = \sup_{t \geq 0} \int_0^t |\bar{g}_{ij}(\rho)| \, \mathrm{d}\rho = \int_0^\infty |\bar{g}_{ij}(\rho)| \, \mathrm{d}\rho.$$

Input–output stability: LTI case

Therefore, we can restate our prior stability observations as the equivalence between the following two statements:

1. A LTI system is uniformly BIBO stable.
2. D is finite and for every entry $\bar{g}_{ij}(\rho)$ of $Ce^{A\rho}B$, we have

$$\int_0^\infty |\bar{g}_{ij}(\rho)| \, \mathrm{d}\rho < \infty.$$

Frequency-domain conditions for BIBO stability

Taking Laplace transforms with respect to temporal variable ρ,

$$\mathcal{L}[\bar{g}_{ij}(\rho)] = \mathcal{L}[Ce^{A\rho}B] = C(sI - A)^{-1}B.$$

$$G_{ij}(s) = \frac{\alpha_0 s^q + \alpha_1 s^{q-1} + \cdots + \alpha_{q-1}s + \alpha_q}{(s-\lambda_1)^{m_1}(s-\lambda_2)^{m_2}\cdots(s-\lambda_k)^{m_k}}$$

Frequency-domain conditions for BIBO stability

Use partial-fraction expansion to invert $G_{ij}(s)$, we would find the intermediate result

$$G_{ij}(s) = \frac{a_{11}}{s-\lambda_1} + \frac{a_{12}}{(s-\lambda_1)^2} + \cdots + \frac{a_{1m_1}}{(s-\lambda_1)^{m_1}} + \cdots$$

$$\frac{a_{k1}}{s-\lambda_k} + \frac{a_{k2}}{(s-\lambda_k)^2} + \cdots + \frac{a_{km_k}}{(s-\lambda_k)^{m_k}}.$$

Frequency-domain conditions for BIBO stability

The inverse Laplace transform is then given by

$$
\begin{aligned}
\bar{g}_{ij}(t) &= \mathcal{L}^{-1}[G_{ij}(s)] \\
&= a_{11}e^{\lambda_1 t} + a_{12}te^{\lambda_1 t} + \cdots a_{1m_1}t^{m_1-1}e^{\lambda_1 t} + \cdots \\
&\qquad a_{k1}e^{\lambda_k t} + a_{k2}te^{\lambda_k t} + \cdots + a_{km_k}t^{m_k-1}e^{\lambda_k t}.
\end{aligned}
$$

Frequency-domain conditions for BIBO stability

1. If for all $G_{ij}(s)$, all the poles have strictly negative real parts, then $\bar{g}_{ij}(t)$ converges to zero exponentially quickly and the system is BIBO stable.
2. If at least one of the $G_{ij}(s)$ has a pole with zero or positive real part, then $\left|\bar{g}_{ij}(t)\right|$ does not converge to zero and the system is not BIBO stable.

Frequency-domain conditions for BIBO stability

Note that adding a constant finite D to $C(s\mathrm{I}-\mathrm{A})^{-1}B$ will not change the poles, we can restate the BIBO stability relationship as

1. The system is uniformly BIBO stable.
2. Every pole of every entry of the transfer function of the system has a strictly negative real part.

BIBO stability vs Lyapunov stability

Example: Consider the example

$$\dot{x}(t) = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 1 \end{bmatrix} x(t).$$

State-transition matrix is $e^{At} = \begin{bmatrix} e^{t} & 0 \\ 0 & e^{-2t} \end{bmatrix}$

which is unbounded and therefore, Lyapunov unstable.

BIBO stability vs Lyapunov stability

However,

$$Ce^{At}B = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} e^{t} & 0 \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = e^{-2t}$$

and therefore the system is BIBO stable.

Continous time Observability

If a system is observable, we can determine the initial condition of the state vector $x(0)$ via processing the input to the system $u(t)$ and the output of the system $y(t)$. Since we can simulate the system if we know $x(0)$ and $u(t)$ this also implies that we can determine $x(t)$ for $t \geq 0$. e.g., for LTI,

Continous time Observability

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)\,\mathrm{d}\tau.$$

■ Consider the LTI SISO system LCCODE

$$\dddot{y}(t) + a_1\ddot{y}(t) + a_2\dot{y}(t) + a_3 y(t) = b_0\dddot{u}(t) + b_1\ddot{u}(t) + b_2\dot{u}(t) + b_3 u(t).$$

If we have a realization of this system in state-space form and we have initial conditions $\mathrm{y}(0), \dot{y}(0), \ddot{y}(0)$,

Continous time Observability

$$y(0) = Cx(0) + Du(0)$$

$$\dot{y}(0) = C(\underbrace{Ax(0) + Bu(0)}_{\dot{x}(0)}) + D\dot{u}(0)$$

$$= CAx(0) + CBu(0) + D\dot{u}(0)$$

$$\ddot{y}(0) = CA^2x(0) + CABu(0) + CB\dot{u}(0) + D\ddot{u}(0).$$

- In general,

$$y^{(k)}(0) = CA^kx(0) + CA^{k-1}Bu(0) + \cdots + CBu^{(k-1)}(0) + Du^{(k)}(0),$$

or,

$$\begin{bmatrix} y(0) \\ \dot{y}(0) \\ \ddot{y}(0) \end{bmatrix} = \underbrace{\begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}}_{\mathcal{O}(C,A)} x(0) + \underbrace{\begin{bmatrix} D & 0 & 0 \\ CB & D & 0 \\ CAB & CB & D \end{bmatrix}}_{\mathcal{T}} \begin{bmatrix} u(0) \\ \dot{u}(0) \\ \ddot{u}(0) \end{bmatrix},$$

Continous time Observability

- Thus, if $\mathcal{O}(C, A)$ is invertible, then

$$x(0) = \mathcal{O}^{-1}\left\{\begin{bmatrix} y(0) \\ \dot{y}(0) \\ \ddot{y}(0) \end{bmatrix} - \mathcal{T}\begin{bmatrix} u(0) \\ \dot{u}(0) \\ \ddot{u}(0) \end{bmatrix}\right\}.$$

- We say that $\{C, A\}$ is an observable pair if $\mathcal{O}$ is nonsingular.

For a SISO system, if $\mathcal{O}$ is nonsingular, then we can determine/estimate the initial state of the system $x(0)$ using only $u(t)$ and $y(t)$ (and therefore, we can estimate $x(t)$ for all $t \geq 0$).

For a MIMO system, if $\mathcal{O}$ is full rank, then we can determine/estimate the initial state of the system $x(0)$ using only $u(t)$ and $y(t)$ (and therefore, we can estimate $x(t)$ for all $t \geq 0$).

Continous time Controllability

Can we generate an input $u(t)$ to set an initial condition quickly?

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t).$$

- If $u(t) = \delta(t)$ and $x(0^-) = 0$, then

$$X(s) = (sI - A)^{-1}BU(s) = (sI - A)^{-1}B.$$

- So, via the Laplace initial-value theorem,

$$\begin{aligned} x(0^+) &= \lim_{s\to\infty} sX(s) \\ &= \lim_{s\to\infty} s(sI - A)^{-1}B \\ &= \lim_{s\to\infty} \left(I - \frac{A}{s}\right)^{-1} B \\ &= B. \end{aligned}$$

Continous time Controllability

- Thus, an impulse input brings the state to B from 0.
- What if $u(t) = \delta^{(k)}(t)$?
- Then

$$X(s) = (sI - A)^{-1} B s^k = \frac{1}{s}\left(I - \frac{A}{s}\right)^{-1} B s^k$$

$$= \underbrace{\frac{1}{s}\left(I + \frac{A}{s} + \frac{A^2}{s^2} + \dots\right)}_{\text{holds for large } s} B s^k$$

$$= B s^{k-1} + A B s^{k-2} + A^2 B s^{k-3} + \dots + \frac{A^k B}{s} + \frac{A^{k+1} B}{s^2} + \cdots$$

- The first terms are impulsive: they have zero value for $t > 0$.

Continous time Controllability

- Thus,

$$x(0^+) = \lim_{s\to\infty} s\left(\frac{A^k B}{s} + \frac{A^{k+1}B}{s^2} + \cdots\right)$$
$$= A^k B.$$

So, if $u(t) = \delta^{(k)}(t)$ then $x(0^+) = A^k B$.

- Now, consider the input

$$u(t) = g_1\delta(t) + g_2\dot{\delta}(t) + \cdots g_n\delta^{(n)}(t).$$

Since $x(0^-) = 0$, $x(0^+) = g_1 B + g_2 AB + \cdots + g_n A^{n-1}B$, or

$$x(0^+) = \underbrace{\begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix}}_{\mathcal{C}} \begin{bmatrix} g_1 \\ g_2 \\ g_3 \end{bmatrix}$$

where $\mathcal{C}$ is called the “controllability matrix.”

Continous time Controllability

For a SISO system, if $\mathcal{C}$ is nonsingular, then there is an impulsive input u such that $x(0^+)$ is any desired vector if $x(0^-) = 0$.

For a MIMO system, if $\mathcal{C}$ is full rank, then there is an impulsive input u such that $x(0^+)$ is any desired vector if $x(0^-) = 0$.

If C is nonsingular, we say $\{A, B\}$ is a controllable pair and the system is controllable.

Continous time Controllability

Example: Controllability canonical form

$$\dot{x}(t) = \begin{bmatrix} 0 & 0 & -a_3 \\ 1 & 0 & -a_2 \\ 0 & 1 & -a_1 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} \beta_1 & \beta_2 & \beta_3 \end{bmatrix} x(t).$$

$$\begin{aligned} \mathcal{C} &= [B \quad AB \quad \cdots \quad A^{n-1}B] \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_n. \end{aligned}$$

Continous time Controllability

- If a system is controllable, we can instantaneously move the state from any known state to any other state, using impulse-like inputs.
- Smooth inputs can effect the state transfer (not instantaneously, though!).

DUALITY: $\{A, B, C, D\}$ controllable $\Longleftrightarrow$ $\{A^T, C^T, B^T, D^T\}$ observable.